

Kristine Yoonseo Lee

yskristine@snu.ac.kr | <https://kristineyslee.github.io/>

RESEARCH INTERESTS

Passionate about developing novel robotic systems that address real-world challenges and enhance human capabilities. My research interests span soft, swarm, and bio-inspired robotics, with a focus on how material and structural mechanics give rise to robotic function, from individual mechanisms to collective behavior.

EDUCATION

- Seoul National University** Mar. 2022 - Feb. 2027
B.S. in Mechanical Engineering | GPA: 3.91/4.00
Seoul, South Korea
- Purdue University** Jan. 2025 - May. 2025
Exchange Student, Department of Mechanical Engineering | GPA: 4.00/4.00
West Lafayette, IN, USA
- Gyeonggi Science High School for the Gifted** Mar. 2019 - Feb. 2022
Major in Physics | GPA: 4.17/4.30
Suwon, South Korea

JOURNAL PUBLICATIONS

J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [J.1] Jaemin Eom, Sung Yol Yu, Woongbae Kim, Chunghoon Park, Kristine Yoonseo Lee, Kyu-Jin Cho. (2024). **MOGrip: Gripper for multiobject grasping in pick-and-place tasks using translational movements of fingers.** *Science Robotics*, Vol. 9, Issue 97. DOI: 10.1126/scirobotics.ado3939

PRESENTATIONS

P=POSTER, O=ORAL

- [P.1] Kristine Yoonseo Lee, Kwanwoo Kim, Ho-Young Kim. (2026). **Spatial Distribution of Flexibility Controls Active Chain Locomotion through Confined Lattices.** Poster to be presented at *IEEE MIT Undergraduate Research Technology Conference (URTC)*, Oct. 11, 2026, Cambridge, MA, USA. (Accepted)

WORK EXPERIENCE

- SNU Microfluids & Soft Matter Laboratory | Prof. Ho-Young Kim** Sep. 2025 - Present
Research Intern
Seoul, South Korea
 - Developed programmable morphing textiles using coiled yarn actuators
 - Integrated coiled yarn actuators into a tensegrity structure
 - Investigated the locomotion of link-bots in porous environments
 - Identified emergent collective behaviors in multi link-bot systems
- Purdue Laboratory for Advanced Multiscale Manufacturing | Prof. Martin Byung-Guk Jun** May. 2025 - Aug. 2025
Summer Undergraduate Research Fellowship (SURF)
West Lafayette, USA
 - Developed path planning algorithms for a 6-DOF robotic arm
 - Designed a heuristic cost function for A* algorithm, generating safe and optimal trajectories
- SNU Biorobotics Laboratory | Prof. Kyu-Jin Cho** Jan. 2024 - Dec. 2024
Undergraduate Researcher (UROP)
Seoul, South Korea
 - Fabricated a tendon-driven underactuated gripper
 - Developed an OpenCV-based vision system for object detection, gripper positioning, and grasping-performance evaluation
 - Analyzed multi-object grasping motions of the human hand using marker-based tracking and extracted dominant motion modes through eigenvector analysis
- Samsung Electronics Device Solutions | Equipment Technology Research Institute** Jul. 2024 - Aug. 2024
Samsung Talent Internship Program
Hwaseong, South Korea
 - Developed a conceptual design for an in-pipe cleaning robot used in semiconductor fabrication facilities
 - Designed a novel caster mechanism that enables smooth locomotion through pipes with varying geometries and conditions

SELECTED HONORS AND AWARDS

- (Applicant) **NSF Graduate Research Fellowship Program (GRFP)** Apr. 2027 (expected)
U.S. National Science Foundation
 - Applicant in the 2027 GRFP competition, award decisions expected in April 2027
- Presidential Science Scholarship of South Korea** Mar. 2022 - Jun. 2026
President of South Korea
 - Presidential scholarship for STEM undergraduate students, Full-ride scholarship

- **Grand Prize | Creative Design Festival, Research Division** Sep. 2026
College of Engineering, Seoul National University
 - Presented research on trapping behaviors of link-bots in porous environments
- **Bronze Prize | Creative Design Festival, Creative Project Division** Sep. 2026
College of Engineering, Seoul National University
 - Designed and built a voice-controlled galvo laser system for pointing to spoken targets on physical surfaces
- **3rd Place | SNU ME AI Robot Challenge** Jul. 2026
Department of Mechanical Engineering, Seoul National University
 - Developed an AI-driven robot capable of vision-based object recognition, grasping, and navigation
- **Grand Prize | Synergy Lab Network** Mar. 2026
Chey Institute for Advanced Studies
 - Proposed a cross-laboratory collaborative research idea, *Neural Response: Latency-Free Hand*
- **2nd Place | Collegiate Rube Goldberg World Championship** Mar. 2025
Rube Goldberg Institute
 - Designed and fabricated a Rube Goldberg system
- **Certificate for the Most Outstanding Completer | Samsung Talent Internship Program** Aug. 2024
People Team, Samsung Electronics Device Solutions
 - Awarded to the participant who completed the Samsung Talent Internship Program with the highest distinction
- **2nd Runner Up | ROBOCON International Design Contest 2023 (Hybrid)** Aug. 2023
Chulalongkorn University
 - Controlled a robotic arm and tuned its motion parameters in a Gazebo simulation
- **Creative Design Award | ROBOCON Contest in Creative Engineering Design class** Nov. 2022
Department of Mechanical Engineering, Seoul National University
 - Designed and fabricated robots for multi-object grasping and goal-oriented delivery
 - Selected as a Korean national representative for online participation in the international ROBOCON

SKILLS

- **Design & Prototyping:** Autodesk Fusion360, Solidworks, AutoCAD, CREO
- **Programming:** C/C++, Python, MATLAB
- **Robotics & Control:** ROS, Arduino
- **Simulation & Analysis:** Ansys, Hyperworks

TEACHING

- **SPLIT (Self-Paced Learn·Ing & Tutoring), SNU** Jan. 2024 - Feb. 2024
Physics Tutor
 - Provided online physics tutoring for 10 incoming SNU engineering students for 6 weeks
 - Graded weekly assignments, provided feedback and Q&A support via email and group chat

LEADERSHIP EXPERIENCE & ACTIVITIES

- **STEM (SNU Engineering Honor Society), SNU** Mar. 2024 - Present
Academic Activity Division Director
 - Organized study-abroad and academic seminars for club members
 - Planned career seminars featuring experienced engineers from industry
- **Purdue American Society of Mechanical Engineers** Jan. 2025 - May. 2025
Member of the Rube Goldberg Team
- **SNU RTY (DIY Car Club)** Mar. 2022 - Dec. 2023
Member of the Mechanical Engineering Team
- **Purdue Symphony Orchestra** Jan. 2025 - May. 2025
Principal 2nd Violin
- **SNU Philharmonic Orchestra** Sep. 2022 - Mar. 2024
1st Violin

ADDITIONAL INFORMATION

Languages: English (Native level), Korean (Native level)